cAiGED

Multimodal Edge-AI Guitar Chord Verification System

Revised Project Abstract & Engineering Baseline — v2

Complete revision of the previous baseline incorporating the latest hardware-reference corrections and scope decisions.

Item	Revised baseline
Application	Embedded verification of predefined guitar chord performances
Initial scope	Five predefined exercises: C, A, G, E and D
Sensing	MIPI CSI-2 camera + microphone/audio
AI	Independent visual and acoustic models
Fusion	Initial deterministic late fusion
MCU	STM32N657X0H3Q
Development board	NUCLEO-N657X0-Q
Primary MCU/reference study	MB1940
External-memory reference	STM32N6570-DK PSRAM implementation
Custom PCB	KiCad
Primary emphasis	High-performance embedded PCB engineering

Baseline rule: no external-memory device, Ethernet PHY, camera, microphone, display, power architecture or AI model is frozen until supported by requirements, documentation and/or measurements.

1. Project Abstract

cAiGED is a custom embedded Edge-AI platform designed to investigate whether visual and acoustic information can be combined to verify predefined guitar chord performances.

The system uses an STM32N657X0H3Q to acquire visual information from a MIPI CSI-2 camera and acoustic information from a microphone. The two modalities are processed independently and then combined to determine whether the observed fingering and resulting sound are consistent with a predefined target chord.

The initial application is deliberately constrained to five predefined guitar exercises associated with the CAGED system: C, A, G, E and D. The system is not initially intended to recognize arbitrary chords, perform general guitar transcription, or provide a complete automated guitar tutor.

Central research question: Can an embedded multimodal AI system determine whether an observed guitar fingering and resulting acoustic output are consistent with a predefined target chord?

The custom PCB is a major engineering objective. It provides the STM32N6 core, sensing interfaces, required storage and display, and the high-speed PCB infrastructure needed to demonstrate BGA escape, controlled impedance, signal integrity, power integrity and hardware validation.

2. Project Scope and Boundaries

The first implementation is intentionally limited to one guitar, a controlled experimental setup, five predefined target exercises, controlled camera geometry, controlled acoustic acquisition, independent visual and audio inference, and a local PASS/FAIL/UNCERTAIN result.

Explicitly outside the initial scope are arbitrary chord recognition, complete guitar transcription, polyphonic transcription of arbitrary playing, general-purpose guitarist assessment, a large language model or PC-class AI agent, billion-parameter neural networks, and use of PC DDR3/DDR3L SO-DIMM modules as STM32N6 system RAM.

This scope reduction is deliberate. It makes the research question experimentally tractable while preserving the multimodal and embedded-AI character of the project.

3. Corrected System Concept

The system is a consistency verifier rather than a generic chord classifier. A target exercise is selected; the camera observes the physical fingering; the microphone observes the resulting sound; the two models produce independent evidence; and a fusion layer compares that evidence with the target.

Target: predefined chord/fingering configuration.

Visual observation: observed fretboard/finger configuration.

Acoustic observation: resulting sounding chord.

Fusion: consistency analysis.

Decision: PASS / FAIL / UNCERTAIN.

Feedback: OLED/display.

A visual D and an acoustic D do not automatically prove that the intended fingering was used. Conversely, a different fingering can sometimes produce the intended musical result. The research therefore concerns consistency between a predefined physical configuration and the acoustic outcome.

4. Research Question and Verification Logic

The two modalities provide complementary information. Visual AI can assess whether the fingers appear to match the predefined configuration, while audio AI can assess whether the resulting sound is consistent with the target.

Visual evidence	Audio evidence	Initial interpretation
Correct/high confidence	Correct/high confidence	Likely successful performance
Incorrect/high confidence	Incorrect/high confidence	Likely incorrect performance
Correct/high confidence	Incorrect/high confidence	Possible acoustic/articulation/muting issue

Visual evidence	Audio evidence	Initial interpretation
Uncertain	Correct/high confidence	Acoustically successful but visually ambiguous
Correct/high confidence	Uncertain	Visually plausible but acoustically unresolved

5. Target Hardware Architecture

The STM32N657X0H3Q remains the central processing device. The custom board is derived from the official STM32N6 reference architecture but contains only circuitry justified by cAiGED requirements.

Block	Status	Decision
STM32N657X0H3Q	Required	Core compute/control
MIPI camera	Required	Visual sensing
Microphone/audio	Required	Acoustic sensing
External nonvolatile memory	Required	Standalone persistent system
External RAM	TBD	Only if memory/bandwidth budget requires it
OLED/display	Required	Local feedback
microSD	Optional	Removable storage
USB	TBD/useful	Define concrete workflow
Ethernet	TBD/optional	Keep only if justified
Temperature sensor	Remove	Not required by research question
Integrated ST-LINK	Remove from final PCB	Use external debug access

6. STM32N657X0H3Q

The obsolete STM32H755 direction is removed. The selected MCU is STM32N657X0H3Q in the VFBGA264 package.

Relevant capabilities for this project include an Arm Cortex-M55 up to 800 MHz, 4.2 MB contiguous SRAM, Neural-ART acceleration, MIPI CSI-2 camera connectivity, image-processing capability, external-memory interfaces, USB and Ethernet capability.

The MCU should not be treated as a miniature PC. AI models, buffers and firmware must be designed around embedded memory, compute and real-time constraints.

7. Memory Architecture — Critical Correction

The original proposal correctly identified memory as important but did not distinguish STM32-compatible embedded memories from PC memory modules.

The user's DDR3/DDR3L SO-DIMM RAM sticks are excluded from the cAiGED design. They are PC memory modules and are not direct STM32N6 external-RAM components.

STM32N6 external RAM must instead use a compatible embedded-memory technology and device selected for the actual interface, electrical requirements, bandwidth and memory budget.

External-memory reference study: MB1940 does not contain an external RAM implementation and therefore is not the reference for this part of the design. The primary reference for external memory is the **STM32N6570-DK**, whose implemented memory architecture includes a **256-Mbit Hexadeca-SPI PSRAM connected through the STM32N6 XSPI interface**.

The correct sequence is: **model and buffer requirements** → **memory budget** → **bandwidth requirement** → **memory technology** → **device** → **PCB architecture.**

External RAM is therefore **TBD**. It may be required for frame buffers, tensors or working memory, but this must be demonstrated by a quantitative budget.

External nonvolatile memory is a separate matter. The standalone system requires an appropriate persistent boot/storage architecture; exact Flash device, density and interface remain TBD.

8. Memory Classes

TCM is intended for latency-sensitive code and data. Internal SRAM can support application buffers and smaller AI data. External RAM, if selected, can support larger frame/tensor/working buffers. External Flash stores firmware, models and assets. microSD, if retained, is removable storage for datasets, recordings and logs.

For external RAM specifically, the STM32N6570-DK PSRAM implementation is the primary practical reference. MB1940 remains the primary reference for the MCU/Nucleo architecture and other circuits it actually implements, but it is not an external-RAM reference.

9. Camera Subsystem

The camera is fundamental because cAiGED requires visual evidence of physical fingering. The first implementation should use a fixed or tightly controlled geometry focused on the relevant fretboard region.

Sensor, resolution, frame rate, lens, field of view, connector and power remain TBD. The initial experiment should not attempt arbitrary-guitarist scene understanding.

10. Visual AI

The visual architecture remains open. The recommended first approach is a compact classifier operating on a controlled fretboard ROI and distinguishing the five predefined configurations. A more structured approach—fretboard localization, finger/keypoint detection, string/fret occupancy and chord interpretation—can be investigated later if it provides meaningful diagnostic value.

The smallest model that meets the required accuracy and latency should be preferred.

11. Audio Subsystem

The audio path remains an independent sensing modality. The exact architecture is not frozen. Candidate implementations include a digital MEMS microphone, an analog microphone with ADC, or an audio codec/SAI/I²S solution.

Selection must consider signal quality, noise, sampling rate, bandwidth, latency, power, PCB complexity and STM32N6 interface availability.

12. Audio AI

The initial acoustic task is constrained classification among C, A, G, E, D and an unknown/uncertain class.

Candidate representations include waveform, STFT, spectrogram, mel-spectrogram and chroma-like features. The representation and model must be selected experimentally.

The first dataset should use a controlled microphone position and controlled playing procedure before introducing broader environmental variation.

13. Multimodal Fusion

The visual and audio paths remain independent until after inference. The first fusion implementation should be ordinary firmware logic using the target, classes and confidence values.

A neural fusion model is optional future work. It is not a hardware requirement and should not be assumed to consume Neural-ART resources.

14. Synchronization

Synchronization is a genuine research problem. Finger placement occurs before the acoustic event, and the sound persists after the strum. The system therefore needs an observation window rather than assuming a camera frame and audio sample are simultaneous.

A practical first implementation can associate an acoustic playing event with the most relevant recent visual observation. Exact timing should be validated experimentally.

15. CAGED Learning Interaction

The initial exercise sequence remains **C** → **A** → **G** → **E** → **D**.

The system selects a target, waits for the player to form and play the chord, acquires visual and acoustic evidence, performs fusion, and reports PASS, FAIL or UNCERTAIN.

The tutor behavior is a demonstration layer and should not expand the hardware scope.

16. Ethernet — Critical Correction

The original document conflated the Nucleo Ethernet implementation with the project's Gigabit Ethernet concept. They are different architectures.

MB1940/Nucleo Ethernet uses an external 10/100 PHY and RMII. It is not a Gigabit Ethernet reference implementation.

If Gigabit Ethernet is retained, the custom board requires a Gigabit-capable PHY and the appropriate STM32N6 RGMII architecture. This cannot be copied from the MB1940 RMII section.

Pin-sharing note: STM32N6 RGMII Ethernet signals use MCU pins that are also associated with certain display and parallel-camera alternate functions. This does **not** conflict with the planned MIPI CSI-2 camera, whose dedicated D-PHY pins are separate. It becomes relevant only if a future revision adds an RGB/LTDC display or a parallel DCMI/PSSI camera.

Ethernet is not required for chord verification. It should be retained only if its connectivity or high-speed PCB-design value justifies the added complexity.

17. USB

USB remains useful but its role is not yet frozen. Possible roles include development, PC communication, data transfer, configuration and diagnostics.

USB is not required for the core multimodal inference concept and should not determine the architecture until a concrete workflow is defined.

18. microSD

microSD is removable storage, not primary AI memory. It may store datasets, recordings, logs, configuration and captured test data.

The AI runtime should use the appropriate internal/external memory architecture rather than treating microSD as high-bandwidth inference memory.

19. OLED and User Interface

A small OLED/display remains useful for immediate local feedback. The minimum UI should show the target and PASS/FAIL/UNCERTAIN result, with detected classes or confidence information if useful.

Display model, resolution, interface and refresh rate remain TBD.

20. Development and Hardware Reference Strategy

NUCLEO-N657X0-Q remains the primary development platform. MB1940 remains the primary reference for the STM32N6 Nucleo architecture and for the circuits it actually implements.

External memory is the exception: MB1940 contains no external RAM circuitry, so it is not used as the memory reference. The **STM32N6570-DK is the primary reference for external memory**, specifically its 256-Mbit Hexadeca-SPI PSRAM over XSPI.

The intended process is: study the appropriate official reference circuit; understand its function; verify it against STM32N6 documentation; reconstruct relevant circuitry in KiCad; remove Nucleo-specific infrastructure; adapt memory, power and interfaces; add cAiGED-specific peripherals; perform electrical review; then lay out the PCB.

The STM32N6570-DK may also be used as a secondary reference for its implemented Flash, camera, audio or Gigabit Ethernet architecture when relevant.

21. Nucleo-Specific Circuitry

Integrated ST-LINK, Arduino headers, Morpho headers, development-board user LEDs/buttons and board-specific connector infrastructure are not automatically part of cAiGED.

The custom board must nevertheless retain appropriate programming/debug access, reset, boot/recovery access and test points.

MCU power, boot, reset, external Flash and required peripheral circuitry should be retained or adapted only after verification.

22. KiCad Migration

Migration from the ST reference design is an engineering verification task, not a one-click import.

Each imported/reconstructed section must be checked against the official schematic, symbols, footprints, net names, values, power domains, pin assignments and design rules. Every deviation from MB1940 should have a documented reason. For external memory, the corresponding STM32N6570-DK implementation is the reference instead.

23. Power Architecture

The final power tree remains open. MB1940 is a reference for MCU power domains, decoupling and implementation, not a component list to copy.

A complete rail and current budget must precede regulator selection. The budget must cover MCU, Flash, external RAM if used, camera, audio, display, USB and Ethernet if retained, together with sequencing and protection requirements.

24. Clock, Reset and Boot

Clock, reset and boot are essential infrastructure. The final architecture must be derived from STM32N6 documentation and checked against MB1940.

Required work includes system clock source, peripheral clocks, NRST, boot configuration, debug access and recovery strategy.

25. PCB Design Objective

The custom PCB is a principal project deliverable. It is expected to demonstrate professional high-performance embedded design.

Key areas are the 264-ball 0.8-mm BGA, BGA escape, external-memory routing if used, XSPI, MIPI CSI-2, USB, Ethernet if retained, power distribution, clocks, audio, controlled impedance, SI/PI and manufacturability.

26. Stack-Up and High-Speed Routing

The PCB stack-up must be selected from the actual fabricator capabilities. Controlled-impedance dimensions must be calculated from the real stack-up rather than generic track-width rules.

MIPI CSI-2 requires differential impedance, skew and return-path control. XSPI and external memory require interface-specific timing and routing constraints. USB and Ethernet require appropriate differential routing and return paths. BGA escape and via technology must be considered before layout freeze.

27. Signal Integrity, Power Integrity and EMC

High-speed interfaces must maintain appropriate reference-plane continuity, controlled impedance where required, short return paths and sensible separation.

Audio requires attention to grounding, supply noise, digital coupling and microphone placement. EMC/EMI must be considered during schematic and placement, not added after routing.

28. Debug and Testability

The final board does not need to reproduce the entire ST-LINK subsystem. It does need robust programming/debugging and recovery access.

Recommended provisions include debug/programming, reset, boot access, UART/logging where useful, power-rail test points and key peripheral test points.

29. Dataset Strategy

The first dataset should be controlled and small enough to support rapid iteration. Visual data should contain the five predefined configurations. Audio data should contain the corresponding acoustic events using a consistent setup.

After the baseline works, variation can be introduced systematically: additional players, lighting, guitar conditions, microphone position, playing strength and background noise.

Dataset leakage must be controlled. The test set should represent genuinely unseen conditions where possible.

30. AI Model Strategy

The project does not require large models. The objective is to find the smallest model that provides useful performance within STM32N6 constraints.

Important measurements are visual-only accuracy, audio-only accuracy, fused accuracy, confusion matrix, inference latency, model size, RAM footprint and NPU/Cortex-M55 workload.

The obsolete 1B/2B/3B parameter experiment is removed. Swap usage is also removed as a primary embedded-AI metric.

31. STM32N6 AI Deployment

The intended flow remains: dataset → model development → validation → quantization/optimization → ST Edge AI tooling → embedded deployment → on-target profiling.

Useful measurements include model size, memory use, inference duration, NPU/Cortex-M55 activity and power. Neural-ART versus Cortex-M55 execution is a valuable experiment because it directly demonstrates the relevance of the selected MCU.

32. Real-Time Firmware

DMA and concurrency should be used where measurements justify them. FreeRTOS is not a predetermined requirement.

A first firmware implementation can simply perform hardware initialization, target selection, camera acquisition, audio acquisition, preprocessing, independent inference, fusion, decision and OLED update, with storage/communications added if required.

33. Hardware/Software Boundary

The PCB should be informed by software requirements, but the final AI system does not have to be complete before schematic work starts.

A preliminary model and memory budget are sufficient to constrain major hardware decisions. The key dependency is: camera requirements + tensor sizes + audio buffers → memory/bandwidth budget → external-memory decision → hardware architecture.

34. Revised Development Strategy

The project is intentionally PCB-focused. AI work provides the minimum evidence needed to make hardware decisions, while the custom PCB remains the principal engineering deliverable.

35. 26-Week Engineering Roadmap

Minimum AI plumbing check — 3 weeks: prove camera capture, small inference, microphone capture and visible result.

Requirements freeze — 2 weeks: freeze functional and electrical requirements.

MB1940 study — 5 weeks: understand and reconstruct the relevant STM32N6 reference architecture. External memory is studied separately from the STM32N6570-DK PSRAM implementation.

Stack-up planning — 1 week: define fabrication constraints and preliminary impedance strategy.

Schematic design — 7 weeks: create and review the custom KiCad schematic.

PCB layout — 2 weeks: placement, BGA escape and high-speed routing.

SI/PI/DRC review — 2 weeks: verify critical interfaces, power and manufacturability.

Fabrication wait + bring-up preparation — 3 weeks: prepare firmware, tests and manufacturing documentation.

Bring-up + AI integration — 1 week: validate hardware and deploy the minimum multimodal pipeline.

Total planned duration: 26 weeks.

36. Portfolio Focus

The PCB is the principal portfolio proof. The AI system supplies the application and engineering justification for the hardware.

Key portfolio deliverables are the STM32N6 schematic, memory architecture, stack-up and impedance calculations, BGA escape strategy, MIPI/XSPI/USB/Ethernet routing, power architecture, SI/PI/DRC review, manufacturing outputs, bring-up procedure and validation log.

37. Risk Register

External RAM: wrong memory technology or topology can block the board. Mitigation: use the STM32N6570-DK PSRAM architecture as the primary external-memory reference and select the final device only after budgeting.

Ethernet: copying RMII from MB1940 while claiming Gigabit would create an architectural error. Mitigation: use correct RGMII/Gigabit architecture or remove the requirement. Check alternate-function conflicts if LTDC display or parallel camera is later added.

Vision: arbitrary scene recognition can become a major computer-vision project. Mitigation: fixed camera and controlled ROI first.

Audio: a model can learn the recording setup rather than the chord. Mitigation: controlled baseline followed by systematic variation.

Fusion: advanced multimodal ML can expand scope. Mitigation: deterministic late fusion first.

PCB: too many optional peripherals can make the first board unnecessarily risky. Mitigation: keep optional blocks removable until justified.

38. Decisions and Open Items

Decided: cAiGED concept; multimodal camera + audio; C/A/G/E/D initial scope; STM32N657X0H3Q; NUCLEO-N657X0-Q; MB1940 as the primary Nucleo/MCU reference; STM32N6570-DK PSRAM implementation as the primary external-memory reference; KiCad custom PCB; independent visual/audio inference; multimodal fusion; local display.

Required: appropriate external nonvolatile memory for the standalone system; camera; audio acquisition; reset/boot/debug; power architecture.

TBD: external RAM final device/density/bus; camera sensor and geometry; audio architecture; OLED; USB role; Ethernet requirement/speed; power tree; PCB stack-up; visual model; audio model; fusion algorithm; synchronization strategy.

39. Immediate Engineering Tasks

Remove obsolete H755/LLM/1–3B content.

Define the five exact physical chord configurations/voicings.

Prove basic camera capture and one small visual inference on NUCLEO-N657X0-Q.

Prove microphone/audio acquisition and one small audio inference.

Measure approximate model, frame-buffer and audio-buffer memory requirements.

Study the STM32N6570-DK external-memory implementation and use it as the primary reference for selecting the cAiGED PSRAM architecture.

Decide whether external RAM is necessary and, if so, select a compatible device.

Decide whether Ethernet is actually required. If Gigabit remains, use the correct RGMII/Gigabit PHY architecture and verify alternate-function pin allocation.

Define USB's concrete purpose.

Study MB1940 sheet by sheet and classify every circuit as required, optional, Nucleo-specific or redesign.

Build the KiCad schematic architecture and freeze the PCB manufacturer/stack-up before detailed routing.

40. Final Simplified Project Concept

cAiGED is an embedded multimodal guitar-chord verifier.

A guitarist is presented with one of five target exercises: C, A, G, E or D.

A camera observes the predefined fingering configuration. A microphone observes the resulting guitar sound. A visual AI model and an audio AI model operate independently. Their outputs are combined by a simple fusion layer.

The system reports whether the physical fingering and acoustic result are consistent with the target: **PASS, FAIL or UNCERTAIN**.

The system runs locally on an STM32N657X0H3Q. A custom PCB provides the camera, audio, display, external nonvolatile memory and only the additional interfaces justified by the final requirements.

External RAM, Ethernet, USB and microSD are optional architectural blocks unless a real requirement or measurement justifies them. If external RAM is required, its architecture is based on an STM32N6-compatible memory device, with the STM32N6570-DK PSRAM implementation as the primary reference.

41. Final Project Definition

cAiGED — Multimodal Edge-AI Guitar Chord Verification System is an embedded research and engineering project investigating whether visual evidence of guitar fingering and acoustic evidence of the resulting chord can be combined on an STM32N657X0H3Q to verify predefined guitar chord performances.

The initial application uses five predefined CAGED-related exercises. Visual and acoustic inference remain independent before a simple fusion stage. The final demonstrator provides local feedback.

The custom PCB is derived from the official STM32N6 reference architecture and is designed as a serious high-performance embedded PCB, including BGA, high-speed interfaces, memory, power, SI/PI and manufacturability considerations.

External PC DDR3/DDR3L RAM modules are excluded. External RAM, if required, is studied and selected using the STM32N6570-DK PSRAM implementation as the primary reference. Gigabit Ethernet is not inherited from the Nucleo and is retained only if the correct RGMII/Gigabit architecture is justified. Optional peripherals are not frozen until their function is established.

42. Engineering Decision Rule

Do not add a component because the reference board contains it. Do not add a feature because it sounds useful. Add it when a cAiGED requirement, measurement or engineering constraint demonstrates that it is needed.

For every MB1940 circuit, ask: what function does it perform, and does cAiGED require that function?

For every new component, ask: what requirement does it satisfy, and what evidence supports its selection?

For external memory, use the STM32N6570-DK implementation as the primary practical reference rather than assuming MB1940 contains an equivalent circuit.

43. Definition of First Success

A valid first demonstrator does not need to be a universal guitar tutor.

It is sufficient to select C/A/G/E/D, acquire a camera observation, acquire an acoustic event, run compact visual and audio inference, fuse the results, and display PASS/FAIL/UNCERTAIN locally.

Once this works, the project can expand through better models, broader environmental variation, richer diagnostics or more advanced fusion.

44. Source Hierarchy for Future Engineering Work

1. STM32N657X0 datasheet/reference manual — electrical limits, interfaces, pin functions, power and memory capabilities.

2. Official NUCLEO-N657X0-Q / MB1940 design — practical STM32N6 implementation for the circuits actually present on the Nucleo.

3. Official STM32N6570-DK design — primary practical reference for external memory, specifically the 256-Mbit Hexadeca-SPI PSRAM over XSPI; also useful for other directly relevant subsystems.

4. Official ST application notes and design guidelines — implementation and layout guidance.

5. Datasheets/reference designs for selected camera, memory, audio, display, Ethernet PHY and connectors.

6. Experimental measurements — actual AI performance, timing, power, memory use, acoustic performance and camera performance.

45. Final Baseline Statement

The original proposal contained a strong core idea but mixed it with obsolete STM32H755 material, PC-class AI experiments, incompatible RAM assumptions and an incorrect equivalence between the Nucleo's 10/100 RMII Ethernet and Gigabit Ethernet.

The revised project removes those blockers while preserving the central multimodal concept.

The external-memory reference is now explicitly separated from MB1940: MB1940 is the Nucleo/MCU reference, while the STM32N6570-DK PSRAM implementation is the primary reference for external RAM.

The Gigabit Ethernet pin-sharing issue is also explicitly bounded: it does not conflict with the planned MIPI CSI-2 camera or small SPI/I²C OLED, but must be checked before adding an LTDC/RGB display or parallel camera.

The resulting project is a constrained guitar-chord verification system on STM32N6 plus a serious custom PCB engineering exercise. That is the baseline to use for subsequent requirements, reference-design analysis, schematic and PCB work.